Fusion Data Card

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Card ID

Primary Dataset	*“On_Time_Reporting_Carrier_On_Time_Performance_(1987_present)_2026_6.csv”*
Description	*This dataset contains 607,577 US domestic flight records reported by 12 airlines across 351 airports, with scheduled and actual times, delays by cause, cancellations and diversions, from 06/01/2026 to 06/30/2026.*
Supplemental Dataset 1	*“nldn-tiles-202606.csv”*
Description	*This dataset contains 475,343 daily lightning-strike counts on a 0.1° latitude/longitude grid covering the continental US, totalling 4,538,490 strikes, from 06/01/2026 to 06/30/2026.*
Supplemental Dataset 2	*“hail-tiles-202606.csv”*

Description	*This dataset contains 338,241 daily radar-detected hail counts on the same 0.1° grid, totalling 2,428,076 detections, from 06/01/2026 to 06/30/2026.*
Supplemental Dataset 3	*“airports.csv”*
Description	*This dataset contains 86,135 airports, heliports and airfields worldwide from OurAirports, with their latitude, longitude and IATA code. We use it only to give each BTS airport a location, because the flight data has no coordinates.*

Datasets at a glance

Primary dataset

The primary dataset contains 607,577 examples representing individual US domestic flight segments (One row per flight) in the period from 06/01/2026 to 06/30/2026. It covers 12 reporting carriers and 351 origin and destination airports. In this period 1.65% of flights were cancelled and 0.39% diverted, with a mean departure delay of 20.7 minutes.

Fields (if applicable)

Field name	Data Type	Description
FlightDate	date (string)	Flight date, YYYY-MM-DD. 2026-06-01 to 2026-06-30. e.g. 2026-06-16
Reporting_Airline	string	Carrier code. 12 values: AA, AS, B6, DL, F9, G4, MQ, OH, OO, UA, WN, YX
Origin / Dest	string	Origin/destination airport code. 351 distinct. e.g. MKE, DTW
CRSDepTime	integer	Scheduled departure, local hhmm. 1–2359. e.g. 1649
DepTime	integer	Actual departure, local hhmm. 1–2400. Null if cancelled

Field name	Data Type	Description
DepDelayMinutes	float	Departure delay, minutes; negative values floored to 0. 0–2759
ArrDelayMinutes	float	Arrival delay, minutes. 0–2752
Cancelled	float (0/1)	1 if cancelled. 10,019 flights (1.65%)
CancellationCode	string	Reason: A=Carrier, B=Weather, C=National Air System, D=Security
Diverted	float (0/1)	1 if diverted. 2,363 flights (0.39%)
WeatherDelay	float	Minutes attributed to weather. 0–1823
NASDelay	float	Minutes attributed to the National Air System. 0–1310
CarrierDelay	float	Minutes attributed to the carrier. 0–2692
LateAircraftDelay	float	Minutes from a late inbound aircraft. 0–2398
Distance	float	Great-circle distance, miles. 31–4983

Supplemental dataset 1

The supplemental dataset contains 475,343 examples representing one grid-cell-day of lightning activity each, across 129,733 distinct 0.1-degree cells spanning longitude −125.6 to −44.5 and latitude 16.6 to 51.7, in the period from 06/01/2026 to 06/30/2026.

Fields (if applicable)

Field name	Data Type	Description
ZDAY	date (integer)	UTC day, YYYYMMDD. 20260601–20260630. e.g. 20260601
CENTERLON	float	Grid cell centre longitude, degrees.

Field name	Data Type	Description
		−125.6 to −44.5. e.g. -96.0
CENTERLAT	float	Grid cell centre latitude, degrees. 16.6 to 51.7. e.g. 37.6
TOTAL_COUNT	integer	Lightning strikes in that cell that day. 1–1,481, mean 9.5

Supplemental dataset 2

The supplemental dataset contains 338,241 examples representing one grid-cell-day of radar-detected hail each, across 93,653 cells spanning longitude −167.8 to 147.9 and latitude 10.5 to 67.4, in the period from 06/01/2026 to 06/30/2026.

Fields (if applicable)

Field name	Data Type	Description
ZDAY	date (integer)	UTC day, YYYYMMDD. 20260601–20260630
CENTERLON	float	Grid cell centre longitude, degrees. −167.8 to 147.9. e.g. -79.9
CENTERLAT	float	Grid cell centre latitude, degrees. 10.5 to 67.4. e.g. 33.0
TOTAL_COUNT	integer	Hail detections in that cell that day. 1–228, mean 7.2

Supplemental dataset 3

The supplemental dataset contains 86,135 examples, one per airport, heliport, seaplane base or closed airfield worldwide. Only 9,054 of them have an IATA code, and 351 of those match the BTS origin and destination airports (97 large, 252 medium and 2 small airports). Added in Milestone 1, because the join to the weather grid needs a location for each airport and the BTS data has none.

Fields (if applicable)

Only the fields we use are listed. The file has 19 columns.

Field name	Data Type	Description
iata_code	string	Three-letter IATA code, the same code BTS uses for Origin/Dest. Empty for 77,081 rows. e.g. AUS
name	string	Airport name. e.g. Austin Bergstrom International Airport
type	string	large_airport, medium_airport, small_airport, heliport, seaplane_base, balloonport or closed
latitude_deg	float	Airport latitude, degrees. e.g. 30.1975
longitude_deg	float	Airport longitude, degrees. e.g. -97.6620
iso_country	string	Country code. e.g. US

Additional notes

- hail-tiles-202606.csv misdescribes itself in its own header comment as “the count of Vaisala NLDN lightning strikes” — identical boilerplate to the lightning file. This is a copy-paste error in NOAA’s header generation, present across their hail, structure and lightning tile products. The data are hail detections; only the column names and filename are reliable.
- The five delay-cause fields are populated for only 153,037 of 607,577 rows, because BTS attributes causes only when arrival delay reaches 15 minutes. These nulls are not missing at random, and treating them as zero is a modelling decision rather than a cleaning step.
- The primary CSV’s header row ends in a trailing comma, producing an empty 110th column that pandas names Unnamed: 110.
- Spatial extents differ between the supplemental sets: lightning covers the continental US, while hail extends to Alaska, Pacific and Caribbean radar sites and crosses the antimeridian. National maps need consistent extents.
- OurAirports lists Palm Beach International under its new code DJT, but the June 2026 BTS file still uses PBI. We map DJT to PBI in the notebook.

Usage information

Questions driving the exploration study

1. How does the daily volume of lightning near an airport relate to that airport's mean departure delay?
2. Which of the 351 airports show the strongest weather-delay sensitivity, and does that track their geography?
3. Does BTS's own WeatherDelay attribution agree with independently observed lightning and hail activity on the same date and location?
4. How do the five delay causes shift in composition on days with severe weather versus clear days?
5. Do hail days and lightning days produce different operational responses (Cancellations versus delays)?
6. How far does weather disruption propagate through LateAircraftDelay to airports with no local severe weather?
7. Which carriers absorb weather disruption best, controlling for their route geography?

Sanctioned Use(s)

- BTS On-Time Performance: collected under 14 CFR Part 234 for public reporting of airline service quality, and published to support the DOT's Air Travel Consumer Report. Public analysis and research are explicitly sanctioned uses.
- NOAA SWDI (both supplemental sets): NCEI publishes the Severe Weather Data Inventory as a research and analysis archive of severe-weather records.
- OurAirports: a community-maintained open dataset, published for any use.
- Discouraged: SWDI hail, mesocyclone and tornado-signature products are radar-algorithm output, not verified ground reports — the raw point files label themselves #This file contains experimental data. They should not be treated as confirmed hailfall, nor substituted for NCEI's Storm Events Database where ground truth matters. Coverage is limited to NEXRAD radar range.

Licensing

- BTS: a work of the US federal government, in the public domain under 17 U.S.C. §105. No restrictions on sharing, transformation or reuse. BTS requests citation as the source.
- NOAA SWDI: NOAA open data — free to use and redistribute, with attribution requested for unaltered data.

- The lightning data originates from Vaisala's commercially-operated NLDN. Under NOAA's contract, raw flash and strike-level data are restricted to government and military users. The 0.1-degree daily tile aggregate we use is the publicly releasable derivative — that aggregation is why the tiles product exists. Our file is unrestricted, but strike-level detail cannot be reconstructed from it.
- OurAirports: released into the public domain by its maintainers. No restrictions and no attribution required.

Privacy considerations

No personally identifiable information in any of the four datasets. The BTS data is operational and flight-level: no passenger, crew or booking records. Tail_Number identifies a specific airframe and no anonymisation or redaction was applied or required. The two SWDI datasets are geophysical observations aggregated to a spatial grid, with no human subjects. The OurAirports data describes public facilities only.

Format

- The primary dataset is in a CSV file.
- The first supplemental dataset is in a CSV file.
- The second supplemental dataset is in a CSV file.
- The third supplemental dataset is in a CSV file.
- The processed outputs of Milestone 1 are Parquet files in the bucket's processed/ folder: fused.parquet (one row per airport-day) and flights_clean.parquet (one row per flight).

Source and collection process

Origin

- Primary: Reporting Carrier On-Time Performance (1987–present) table, "On-Time" database, BTS TranStats data library: https://www.transtats.bts.gov/ontime/
- Supplemental 1 and 2: NOAA NCEI Severe Weather Data Inventory, CSV archive: https://www.ncei.noaa.gov/pub/data/swdi/database-csv/v2/ (files nldn-tiles-202606.csv.gz and hail-tiles-202606.csv.gz)
- Supplemental 3: OurAirports open data, https://ourairports.com/data/ (file airports.csv, from the

mirror at https://davidmegginson.github.io/ourairports-data/airports.csv)

Original datasets author/publisher

- Primary: US Department of Transportation, Bureau of Transportation Statistics, Office of Airline Information. Collected under 14 CFR Part 234 from carriers accounting for at least 0.5% of domestic scheduled passenger revenues — which is why only 12 airlines appear.
- Supplemental: NOAA National Centers for Environmental Information (NCEI), Asheville, NC. Lightning observations are sourced from Vaisala Inc.'s National Lightning Detection Network under contract; hail detections derive from the NEXRAD Level-III Hail Detection Algorithm.
- Supplemental 3: OurAirports, a volunteer project started by David Megginson, with edits from its user community.

Original datasets point of contact

- Primary: BTS TranStats support — https://www.transtats.bts.gov/
- Supplemental: NCEI Customer Engagement Branch (ncei.orders@noaa.gov)
- Supplemental 3: OurAirports, https://ourairports.com/

Original metadata

- Primary: readme.html, shipped with the BTS extract; converted and stored at Datasets/Primary/README.md. Field definitions also at https://www.transtats.bts.gov/Fields.asp?gnoyr_VQ=FGJ
- Supplemental: SWDI documentation at https://www.ncei.noaa.gov/pub/data/swdi/database-csv/v2/ and https://www.ncei.noaa.gov/products/severe-weather-data-inventory
- Supplemental 3: data dictionary at https://ourairports.com/help/data-dictionary.html

Collection process

- Primary: self-reported by airlines to DOT monthly under the Airline Service Quality Performance programme; released ~2 months after month end. June 2026 was published 12 August 2026. Details: https://www.bts.gov/topics/airlines-and-airports/understanding-reporting-causes-flight-delays-and-cancellations
- Supplemental: NLDN sensors detect cloud-to-ground flashes continuously; NEXRAD radars run the hail detection algorithm per volume scan. NCEI aggregates both to daily counts per 0.1-degree grid cell on UTC day boundaries.

- Supplemental 3: volunteers add and correct airport records by hand, and the maintainers export the full table as CSV every night.

Collection or access date

Primary dataset downloaded 31 August 2026. Supplemental datasets 1 and 2 downloaded 31 August 2026. Supplemental dataset 3 downloaded 28 September 2026.

Refresh date/schedule

- Primary: monthly, roughly two months in arrears. June 2026 released 12 August 2026; July expected mid-September. The file we hold is a fixed monthly extract and will not change.
- Supplemental: monthly files are posted early in the following month and revised — nldn-tiles-202606 was written 5 July 2026, and NCEI's directory shows August 2026 files updated as recently as 31 August 2026. Our copies are point-in-time snapshots taken 31 August 2026.
- Supplemental 3: updated nightly. Our copy is a snapshot taken 28 September 2026. Airport coordinates almost never change, but codes can: the PBI to DJT rename is an example.

Cleaning and processing

Missing values

We found missing values with isna().sum() on each DataFrame, then checked the cause of each group before we chose how to handle it. The notebook uses assert statements to test each cause against the data, so a wrong assumption stops the notebook.

Field(s)	Missing	Cause	Handling
carrier_delay, weather_delay, nas_delay, security_delay, late_aircraft_delay	454,540 (74.8%)	BTS gives delay causes only for flights that arrive 15+ minutes late. The cause fields are present on every such flight and missing on every other flight.	Filled with 0. A missing cause means that no delay minutes were attributed, not that the value is unknown.

Field(s)	Missing	Cause	Handling
dep_delay, dep_time, dep_del15	9,554 (1.6%)	Every one of these flights was cancelled.	Kept as missing. A 0 would count a cancelled flight as on time and understate storm effects. The cancelled flag records the outcome.
arr_delay, arr_time, arr_del15	12,382 (2.0%)	Exactly the cancelled flights plus the diverted flights.	Kept as missing, for the same reason.
cancellation_code	597,558 (98.4%)	The flight was not cancelled.	Kept as missing. This is correct.
tail_number	832 (0.14%)	Not reported by the airline.	Kept. No step uses this field.
Div3–Div5 fields, Unnamed: 109	100%	Empty columns.	Dropped with the other unused columns.
lightning, hail	0		Nothing to do.
airports.iata_code	77,081 (89.5%)	Heliports, airstrips and closed fields have no IATA code.	Rows without a code are dropped, because they cannot match a BTS airport.
fused: mean_dep_delay, mean_arr_delay	11 airport-days	Small airports where every flight that day was cancelled, so there is no delay to average.	Kept as missing. cancel_rate is 1.0 on these rows.
fused: lightning_count, hail_count	8,283 / 8,297 before fill	An airport-day whose tile has no row in the weather file.	Filled with 0. The tile files list only cells with at least one detection, so no row means no detections.

We also converted types so that missing values are stored correctly: the 0/1 delay flags use the nullable boolean type, and the hhmm clock times use the nullable Int64 type.

Aggregation

The flights are one row per flight, and the weather is one row per 0.1° tile per day. We grouped the flights

by origin airport and date with groupby(['flight_date', 'origin']), so that the target grain is **airport × day**, the finest grain that both sides share. This turned 607,577 flights into 10,306 airport-days.
For each airport-day we compute:

- n_flights: the number of scheduled departures.
- mean_dep_delay, mean_arr_delay: the mean delay in minutes. We use the mean so that busy and quiet airports are comparable. The median would hide the long storm tail that we want to study.
- cancel_rate, divert_rate: the fraction of flights cancelled or diverted.
- carrier_delay, weather_delay, nas_delay, security_delay, late_aircraft_delay: the total delay minutes for each cause, so that the size of a disruption stays visible.

We then filtered the airport-days to the **contiguous US**. We dropped 890 airport-days in Alaska, Hawaii, Puerto Rico, the US Virgin Islands and the Pacific territories, which leaves 9,416 airport-days at 320 airports. The lightning network (NLDN) is built for the lower 48 states, so a zero lightning count outside it would mean “not measured”, not “no lightning”.

We did not smooth the data before the join. The daily mean departure delay is noisy (8.1 to 33.4 minutes, standard deviation 7.2), and a 7-day rolling mean shows its trend clearly. But the day-to-day spikes follow the storms (the daily mean delay has a correlation of 0.68 with the daily lightning count at airport tiles), and those spikes are the signal we want to study. We use smoothing only for trend charts.

Dates: the weather ZDAY is a UTC date, and flight_date is a local date. A UTC day starts at 7pm CDT (8pm EDT, 5pm PDT) the evening before, so a storm late in the local evening counts toward the next day. We accept this at a daily grain: most storms happen in the local afternoon, before midnight UTC, and the tile counts are daily totals that we cannot shift. This mismatch can only weaken the link between storms and delays. It cannot create a false one.

Joins of input sources

The join has two steps:

1. **Airports to coordinates.** A left join of the airport-days to airports.csv on the airport code (origin = iata_code) adds a latitude and longitude to each airport. All 351 BTS airports match after we map DJT back to PBI. An assert stops the notebook if any airport has no coordinates.
2. **Airport-days to weather.** We round each airport’s latitude and longitude to one decimal place. This gives the center of the 0.1° tile that contains the airport (about 11 × 9 km, close to the ~5 mile lightning “ramp closure” radius). Then we left join lightning and hail on **(date, latitude, longitude)**, where flight_date matches ZDAY.

We use left joins so that every airport-day stays in the result, with or without weather. A missing tile row

means no detections, so we fill it with 0 (see Missing values). An assert checks that the join adds no rows, which proves that the weather keys are unique.
The result, fused.parquet, has 9,416 airport-days at 320 airports. 1,133 airport-days (12.0%) have lightning in the airport tile, and 1,119 (11.9%) have hail. A first comparison shows that the join carries a real signal:

	Mean departure delay	Cancellation rate	Weather-delay minutes per airport-day
Lightning in the airport tile	26.8 min	4.1%	319.5
No lightning	16.6 min	1.7%	48.7

A known limit: one tile can miss a storm just outside it. Summing the 3 × 3 block of tiles around each airport is the next thing to try.

Exploration study

Descriptive statistics

Primary dataset (after cleaning, 607,577 flights, 31 fields). The delay-cause fields are after the fill with 0.

Field	Count	Missing	Mean	Std	Min	Median	Max
dep_delay (min)	598,023	9,554	17.62	62.55	−34	−1	2,759
arr_delay (min)	595,195	12,382	11.68	64.51	−73	−5	2,752
distance (miles)	607,577	0	854.03	611.82	31	693	4,983
carrier_delay (min)	607,577	0	5.76	38.46	0	0	2,692
weather_de	607,577	0	1.26	18.31	0	0	1,823

Field	Count	Missing	Mean	Std	Min	Median	Max
lay (min)							
nas_delay (min)	607,577	0	3.80	20.23	0	0	1,310
security_delay (min)	607,577	0	0.02	1.35	0	0	539
late_aircraft_delay (min)	607,577	0	7.87	34.20	0	0	2,398

Field	Type	Distinct values	Most common	Notes
flight_date	datetime	30		2026-06-01 to 2026-06-30
airline	category	12	WN (120,933 flights)	
origin / dest	category	351 / 351	ORD (31,374 departures)	
dep_time_blk	category	19	0700-0759 (45,602)	
cancelled	bool	2	False	1.65% true (10,019)
diverted	bool	2	False	0.39% true (2,363)
dep_del15 / arr_del15	boolean	2	False	26.6% / 25.7% true
cancellation_code	category	4	B = Weather (6,603)	A = Carrier 2,124, C = NAS 1,286, D = Security 6
tail_number	string	5,654	N480HA (286)	832 missing

Delays are strongly right-skewed: the median departure is 1 minute early, but the mean is 17.6 minutes late because of a long tail of large delays. Weather is the most common reason for a cancellation (66% of cancellations with a code), but weather causes only 1.26 minutes of delay per flight on average, much less than late aircraft (7.87) and the carrier (5.76).

Supplemental datasets 1 and 2 (weather tiles).

Field	Count	Missing	Mean	Std	Min	Median	Max	Total
lightning TOTAL_COUNT	475,343	0	9.55	22.69	1	3	1,481	4,538,490
hail TOTAL_COUNT	338,241	0	7.18	9.83	1	4	228	2,428,076

Both tile files cover 30 days and use the same 0.1° grid: every latitude and longitude is a multiple of 0.1, and no (day, latitude, longitude) key repeats.

Supplemental dataset 3 (airports). 86,135 rows: 42,746 small airports, 23,224 heliports, 13,548 closed fields, 4,106 medium airports, 1,273 seaplane bases, 1,175 large airports and 63 balloon ports. 9,054 rows have an IATA code, and the 351 that match BTS are 97 large, 252 medium and 2 small airports.

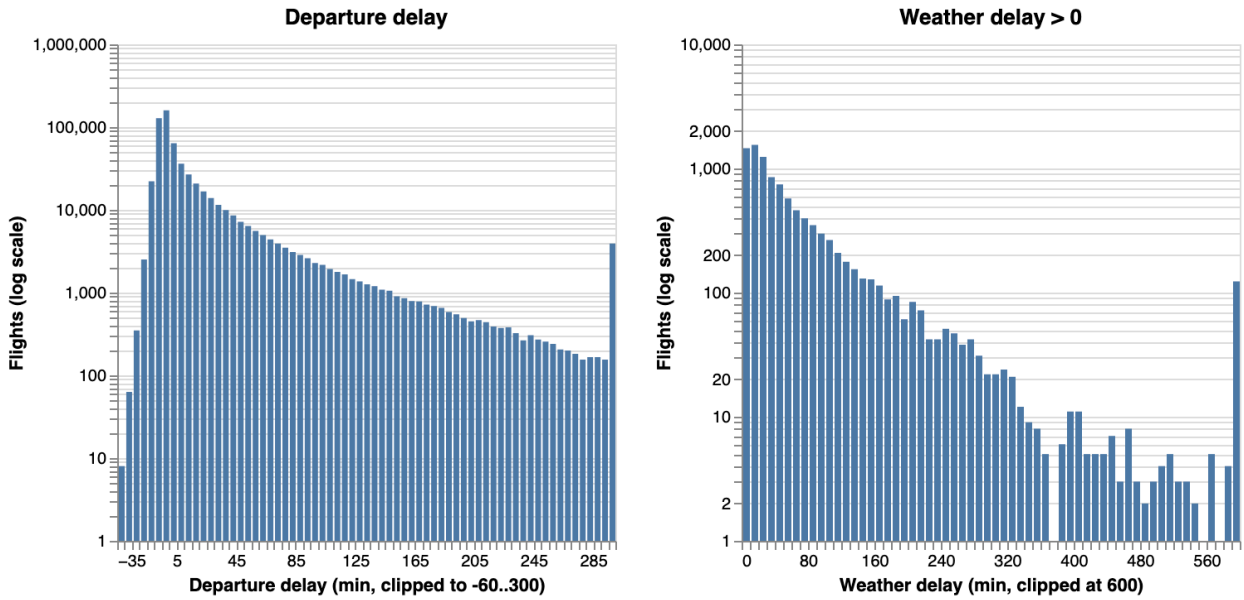
Processed dataset (fused, 9,416 airport-days at 320 airports).

Field	Count	Missing	Mean	Std	Min	Median	Max
n_flights	9,416	0	62.55	143.72	1	10	1,070
mean_dep_delay (min)	9,405	11	17.83	45.42	−23.5	8.56	1,197
mean_arr_delay (min)	9,405	11	12.71	47.25	−43	3.63	1,264
cancel_rate	9,416	0	0.02	0.07	0	0	1
divert_rate	9,416	0	0.00	0.03	0	0	0.5
weather_delay (total min)	9,416	0	81.28	550.98	0	0	22,064
nas_delay (total min)	9,416	0	242.93	765.65	0	17	17,207
late_aircraft	9,416	0	500.31	1,669.25	0	33	47,765

Field	Count	Missing	Mean	Std	Min	Median	Max
_delay (total min)							
lightning_count	9,416	0	1.53	10.31	0	0	307
hail_count	9,416	0	0.96	4.48	0	0	89

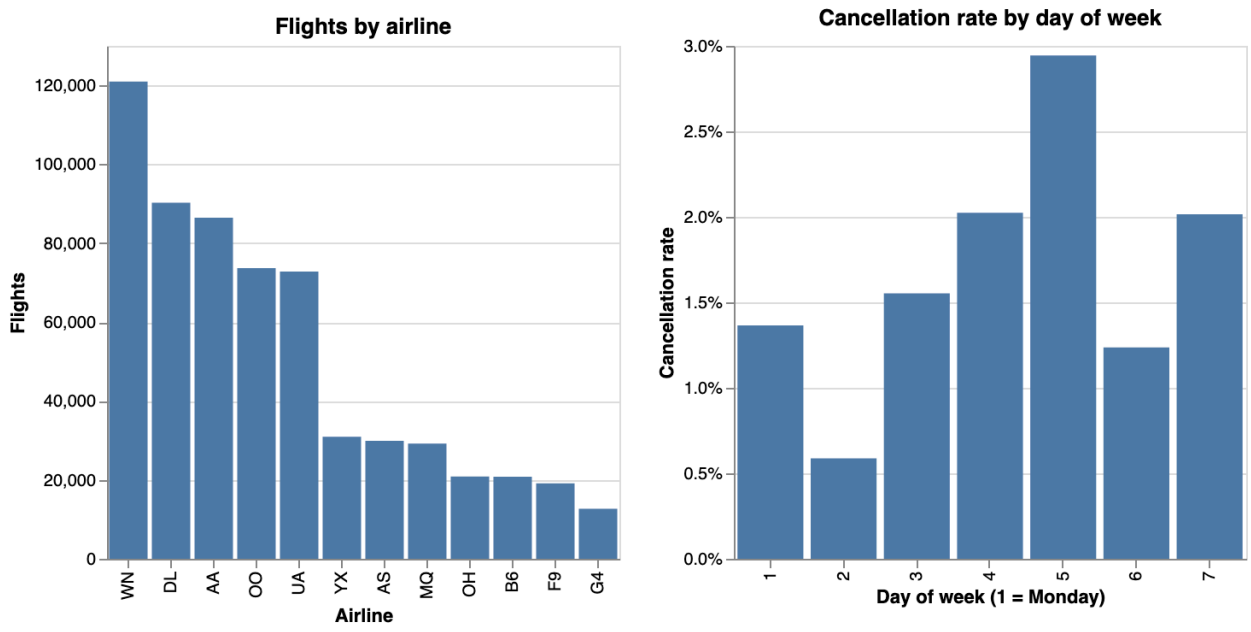
Airport size is very uneven: the median airport-day has 10 departures, but the largest has 1,070. Any comparison across airports must weight by flights or use rates.

Distribution plots



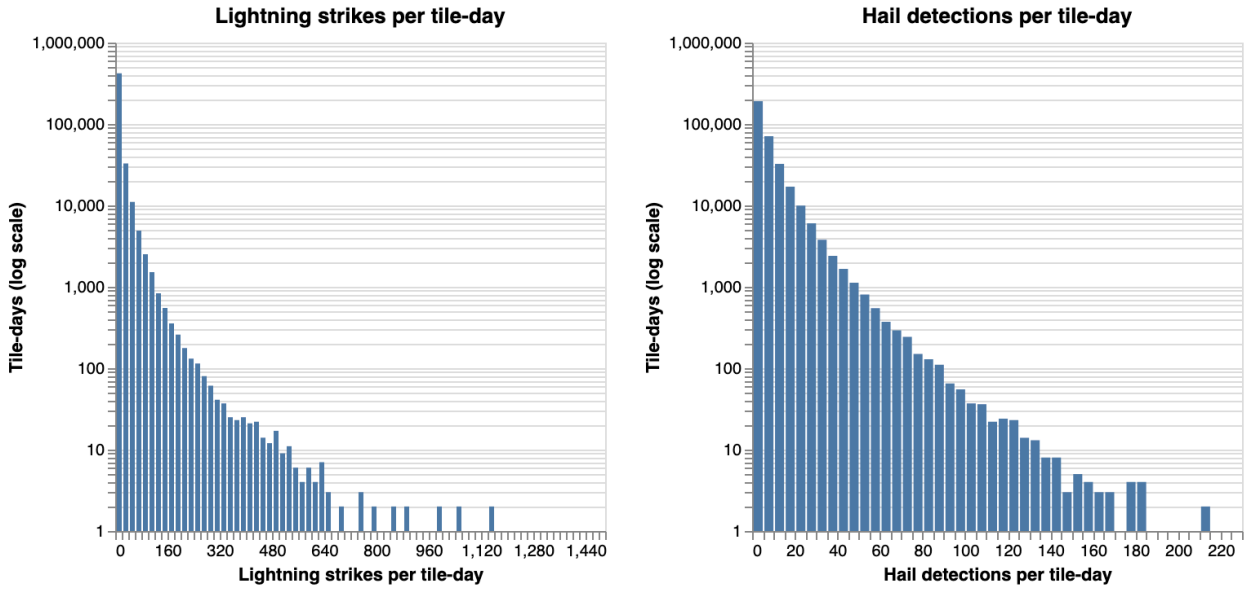
Departure delay (all flights) and weather delay (flights with weather delay > 0). Both use a log scale because of the long right tail.

The departure-delay distribution peaks just before 0 (most flights leave a few minutes early) and falls off slowly to the right. The spike at 300 is every flight delayed 300 minutes or more, collected by the clip. Weather delay is non-zero on 10,127 flights. Its median is 39 minutes and 64% of those delays are an hour or less, but 122 flights waited 10 hours or more.



Flights by airline, and cancellation rate by day of week.

Five airlines (WN, DL, AA, OO, UA) operate 73% of flights. The cancellation rate changes from 0.59% on Tuesdays to 2.94% on Fridays. June has only 4 or 5 of each weekday, so this probably shows which days had storms more than a real weekday effect.



Lightning strikes and hail detections per tile-day, log scale.

Both weather counts are heavily skewed: most tile-days have only a few detections, and a small number of storm cells produce hundreds. This is why we compare “any lightning in the airport tile” against “none”

before we use the raw counts.

Visual analysis

Select and include key charts that best represent the primary insights uncovered during your data exploration.

Streamlit application

Select and include key pages that let users explore your core questions interactively.

Vis Agent samples

Include a few representative visualizations from your vis agent, each with its prompt. Choose examples that show the agent's range; save any assessment for the Agent evaluation study.

Evaluations

First user study

Summarize what users reported learning about the data through your visual analysis. Group recurring themes rather than listing every comment.
List the questions your users still had after exploring the application.
Record the suggestions users offered for making the visualizations clearer or more useful.

Agent evaluation study

Summarize the test questions and your observations of the agent's behavior across runs. Document three areas of improvement grounded in those observations.

Second user study (extra credit)

List the questions users asked your Vis Agent.
Summarize what they learned about the data. Group recurring themes rather than listing every comment.
Record the suggestions users offered for making the agent more functional and useful.